



PAINT EDITOR · BACKDROPS

Create your own maze as a Scratch backdrop



STEPS TO FOLLOW

1. Open Scratch → delete cat
2. Click stage backdrop thumbnail → "Paint" (pencil icon)
3. Rectangle tool with solid dark blue fill
4. Draw thick rectangles for walls
5. Leave 30–40px corridors
6. Draw outer border first → back-arrow to return

TIP

Keep paths wide! If they are too narrow, the hero will get stuck in the walls.

DID YOU KNOW?

In Scratch, "touching colour" checks the exact pixel colour — that's why solid, consistent wall colours are so important.



SPRITES · SIZE

Add a small hero sprite that fits inside the maze corridors



STEPS TO FOLLOW

1. Choose a Sprite → pick small character (Avery or arrow)
2. Set Size to 25–30
3. Drag to maze start
4. Note the x and y coordinates for next step

TIP

After placing the hero, check that it's smaller than the maze corridors. If it looks cramped, reduce the size further.

DID YOU KNOW?

You can click anywhere on the stage while editing to see those x/y coordinates in the bottom-right of the editor.



EVENTS · MOTION · Y-AXIS

Press the up and down arrow keys to move the hero vertically through the corridor



SCRATCH BLOCKS — ON THE HERO SPRITE

when  clicked

go to x: (-190) y: (150)

forever

if <key [up arrow] pressed?> then

change y by (4)

if <key [down arrow] pressed?> then

change y by (-4)

TIP

"change y by (4)" moves UP because

positive y is upward in Scratch.

Negative y moves down — just like a

maths graph!

DID YOU KNOW?

In Scratch, $y=0$ is the centre of the

stage. $y=180$ is the very top; $y=-180$

is the very bottom.



2D VECTORS · 4-WAY INPUT

Add left and right arrow keys to complete full 4-direction movement



SCRATCH BLOCKS — ON THE HERO SPRITE

when clicked

go to x: (-190) y: (150)

forever

if <key [up arrow] pressed?> then

change y by (4)

if <key [down arrow] pressed?> then

change y by (-4)

if <key [left arrow] pressed?> then

change x by (-4)

if <key [right arrow] pressed?> then

change x by (4)

TIP

Use the same speed value (4) for all four directions so movement feels balanced.

DID YOU KNOW?

Vectors power every game engine, physics simulator, animation system, and robotics controller in existence.



COLOUR SENSING · PIXEL DETECTION

Detect when the hero touches a wall — say "Ouch!" as proof it works



SCRATCH BLOCKS — ADD INSIDE THE FOREVER LOOP

(at the end of the forever loop)

```
if <touching color [■ dark blue]?> then
```

```
  say [Ouch! 🍌] for (0.5) secs
```

💡 TIP

The eyedropper only works when you click it first, THEN click the wall on the stage. The colour must match exactly.

🧐 DID YOU KNOW?

"touching color?" reads actual pixel data on screen — making the maze artwork itself the collision system. No coordinate math needed!



ROLLBACK PATTERN · COLOUR SENSING

Reverse each move if the hero touches a wall — the "rollback" pattern



SCRATCH BLOCKS — REPLACE EACH DIRECTION BLOCK

```
if <key [up arrow] pressed?> then
```

```
  change y by (4)
```

```
if <touching color [■ dark blue]?> then
```

```
  change y by (-4) ← rollback!
```

```
... repeat for all 4 directions
```

💡 TIP

Do the EXACT same pattern for all 4 directions — up/down use y, left/right use x. Each direction gets its own immediate rollback.

🧐 DID YOU KNOW?

Database transactions, Git commits, undo/redo stacks, and save files all use rollback — "save state, try action, restore if bad."



MULTIPLE SPRITES · PARALLEL SCRIPTS

Place a spinning star at the maze exit — it runs its own code independently



SCRATCH BLOCKS — ON THE STAR SPRITE

— STAR: when  clicked —

forever

turn (4) degrees

TIP

Drag the star to the very end of your maze path — players must navigate all the way through to reach it.

DID YOU KNOW?

Multiple "when  clicked" scripts run simultaneously. This is called parallel execution — the same concept used in async programming, game engines, and operating systems.



WIN CONDITIONS • BOOLEAN LOGIC

Celebrate when the hero reaches the exit star — stop the game on victory



SCRATCH BLOCKS — ON THE STAR SPRITE (SECOND SCRIPT)

– STAR: when clicked (win script) –

forever

if <touching [Hero]?> then

say [You escaped! 🎉] for (3) seconds

stop [all]

💡 TIP

The star now has TWO scripts — one spinning forever, one watching for the hero. Both run at the same time!

🧠 DID YOU KNOW?

Achievement systems, form validation, access control — all evaluate a boolean condition, then branch to a reward or error path.



VARIABLES • COUNTDOWN TIMER

Add a 30-second countdown; game over if time runs out



SCRATCH BLOCKS — TIMER SCRIPT (ANY SPRITE)

when clicked (timer script)

set [Timer] to (30)

repeat (30)

wait (1) seconds

change [Timer] by (-1)

say [Time's up!] for (2) seconds

stop [all]

TIP

To make it harder, reduce the timer to 20 or 15 seconds. To make it easier, increase it to 60.

DID YOU KNOW?

Login session timeouts, OTP codes, rate limiters, and cooking timers all work exactly like this countdown pattern.